Equivalent Fractions and Mixed Numbers

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS numerator · denominator · equivalent · simplify · improper fraction · mixed number

A Equivalent fractions ●○○ — ANSWER ON THE SHEET

Remember: Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

1. Fill in the box to make the fractions equal.

a) $\frac{2}{3} = \frac{\square}{12}$ **b)** $\frac{3}{5} = \frac{\square}{15}$ **c)** $\frac{1}{4} = \frac{5}{\square}$ **d)** $\frac{5}{6} = \frac{\square}{12}$

e) $\frac{3}{8} = \frac{9}{\square}$ **f)** $\frac{4}{7} = \frac{\square}{35}$ **g)** $\frac{7}{10} = \frac{\square}{100}$ **h)** $\frac{5}{9} = \frac{20}{\square}$

i) $\frac{1}{2} = \frac{\square}{18}$ **j)** $\frac{2}{9} = \frac{12}{\square}$ **k)** $\frac{11}{12} = \frac{\square}{36}$ **l)** $\frac{3}{4} = \frac{\square}{100}$

B Simplifying fractions ●○○ — ANSWER ON THE SHEET

Remember: Divide the top and the bottom by their highest common factor.

2. Write each fraction in its simplest form.

a) $\frac{6}{8}$ = _____ **b)** $\frac{10}{15}$ = _____ **c)** $\frac{12}{18}$ = _____ **d)** $\frac{9}{21}$ = _____

e) $\frac{20}{28}$ = _____ **f)** $\frac{16}{40}$ = _____ **g)** $\frac{27}{45}$ = _____ **h)** $\frac{36}{60}$ = _____

i) $\frac{35}{50}$ = _____ **j)** $\frac{22}{55}$ = _____ **k)** $\frac{42}{56}$ = _____ **l)** $\frac{75}{100}$ = _____

C Improper fractions to mixed numbers ●●○ — ANSWER ON THE SHEET

Remember: Divide the top by the bottom. The remainder becomes the top of the fraction part.

3. Write each improper fraction as a mixed number.

a) $\frac{7}{2}$ = _____ **b)** $\frac{11}{3}$ = _____ **c)** $\frac{17}{5}$ = _____ **d)** $\frac{9}{4}$ = _____

e) $\frac{23}{6}$ = _____ **f)** $\frac{29}{8}$ = _____ **g)** $\frac{31}{10}$ = _____ **h)** $\frac{40}{7}$ = _____

i) $\frac{19}{3}$ = _____ **j)** $\frac{50}{9}$ = _____ **k)** $\frac{47}{12}$ = _____ **l)** $\frac{38}{5}$ = _____

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D Mixed numbers to improper fractions ●●○ — ANSWER ON THE SHEET

> **Remember:** Multiply the whole number by the bottom, then add the top.

4. Write each mixed number as an improper fraction.

a) $2\frac{1}{3}$ = _____ **b)** $1\frac{3}{4}$ = _____ **c)** $3\frac{2}{5}$ = _____ **d)** $4\frac{1}{2}$ = _____

e) $2\frac{5}{6}$ = _____ **f)** $5\frac{3}{8}$ = _____ **g)** $6\frac{2}{3}$ = _____ **h)** $3\frac{7}{10}$ = _____

i) $7\frac{1}{4}$ = _____ **j)** $4\frac{5}{9}$ = _____ **k)** $10\frac{1}{2}$ = _____ **l)** $2\frac{7}{12}$ = _____

E Comparing fractions ●●● — WORK IN YOUR BOOK

> **Remember:** Write both fractions with the same denominator, then compare the tops.

5. Which fraction is larger?

a) $\frac{2}{3}$ or $\frac{3}{5}$ **b)** $\frac{3}{4}$ or $\frac{5}{8}$ **c)** $\frac{1}{2}$ or $\frac{4}{9}$

d) $\frac{5}{6}$ or $\frac{7}{9}$ **e)** $\frac{3}{10}$ or $\frac{1}{4}$ **f)** $\frac{7}{12}$ or $\frac{5}{8}$

g) $\frac{4}{5}$ or $\frac{7}{9}$ **h)** $\frac{2}{7}$ or $\frac{1}{3}$ **i)** $\frac{11}{15}$ or $\frac{7}{10}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$ **b)** $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$ **c)** $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$

d) $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{8}$ **e)** $\frac{1}{3}$, $\frac{2}{9}$, $\frac{5}{18}$ **f)** $\frac{3}{4}$, $\frac{4}{5}$, $\frac{7}{10}$

★ Extension — FOR EARLY FINISHERS

7. Write $\frac{84}{126}$ in its simplest form.

8. Write a fraction that is greater than $\frac{1}{3}$ and less than $\frac{1}{2}$.

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Equivalent Fractions and Mixed Numbers

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS numerator · denominator · equivalent · simplify · improper fraction · mixed number

A Equivalent fractions ●○○

ANSWER ON THE SHEET

Remember: Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

1. Fill in the box to make the fractions equal.

a) $\frac{3}{4} = \frac{\square}{12}$ **b)** $\frac{2}{5} = \frac{\square}{20}$ **c)** $\frac{1}{6} = \frac{5}{\square}$ **d)** $\frac{5}{8} = \frac{\square}{16}$

e) $\frac{2}{7} = \frac{6}{\square}$ **f)** $\frac{4}{9} = \frac{\square}{45}$ **g)** $\frac{3}{10} = \frac{\square}{100}$ **h)** $\frac{5}{6} = \frac{20}{\square}$

i) $\frac{1}{2} = \frac{\square}{14}$ **j)** $\frac{4}{9} = \frac{24}{\square}$ **k)** $\frac{7}{12} = \frac{\square}{36}$ **l)** $\frac{1}{4} = \frac{\square}{100}$

B Simplifying fractions ●○○

ANSWER ON THE SHEET

Remember: Divide the top and the bottom by their highest common factor.

2. Write each fraction in its simplest form.

a) $\frac{4}{6}$ = _____ **b)** $\frac{12}{15}$ = _____ **c)** $\frac{14}{21}$ = _____ **d)** $\frac{8}{20}$ = _____

e) $\frac{18}{30}$ = _____ **f)** $\frac{15}{40}$ = _____ **g)** $\frac{28}{49}$ = _____ **h)** $\frac{48}{72}$ = _____

i) $\frac{45}{60}$ = _____ **j)** $\frac{26}{65}$ = _____ **k)** $\frac{36}{54}$ = _____ **l)** $\frac{60}{100}$ = _____

C Improper fractions to mixed numbers ●●○

ANSWER ON THE SHEET

Remember: Divide the top by the bottom. The remainder becomes the top of the fraction part.

3. Write each improper fraction as a mixed number.

a) $\frac{9}{2}$ = _____ **b)** $\frac{10}{3}$ = _____ **c)** $\frac{13}{5}$ = _____ **d)** $\frac{15}{4}$ = _____

e) $\frac{25}{6}$ = _____ **f)** $\frac{27}{8}$ = _____ **g)** $\frac{43}{10}$ = _____ **h)** $\frac{37}{7}$ = _____

i) $\frac{22}{3}$ = _____ **j)** $\frac{52}{9}$ = _____ **k)** $\frac{41}{12}$ = _____ **l)** $\frac{33}{5}$ = _____

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D Mixed numbers to improper fractions ●●○ — ANSWER ON THE SHEET

> **Remember:** Multiply the whole number by the bottom, then add the top.

4. Write each mixed number as an improper fraction.

a) $3\frac{1}{2}$ = _____ **b)** $1\frac{2}{3}$ = _____ **c)** $2\frac{3}{5}$ = _____ **d)** $4\frac{3}{4}$ = _____

e) $3\frac{1}{6}$ = _____ **f)** $2\frac{5}{8}$ = _____ **g)** $5\frac{1}{3}$ = _____ **h)** $4\frac{9}{10}$ = _____

i) $6\frac{3}{4}$ = _____ **j)** $3\frac{4}{9}$ = _____ **k)** $9\frac{1}{2}$ = _____ **l)** $2\frac{5}{12}$ = _____

E Comparing fractions ●●● — WORK IN YOUR BOOK

> **Remember:** Write both fractions with the same denominator, then compare the tops.

5. Which fraction is larger?

a) $\frac{3}{4}$ or $\frac{2}{3}$ **b)** $\frac{2}{5}$ or $\frac{3}{10}$ **c)** $\frac{1}{2}$ or $\frac{5}{9}$

d) $\frac{5}{6}$ or $\frac{3}{4}$ **e)** $\frac{7}{10}$ or $\frac{3}{4}$ **f)** $\frac{5}{12}$ or $\frac{3}{8}$

g) $\frac{3}{5}$ or $\frac{5}{9}$ **h)** $\frac{3}{7}$ or $\frac{1}{2}$ **i)** $\frac{8}{15}$ or $\frac{3}{5}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{5}{8}$, $\frac{1}{4}$ **b)** $\frac{1}{3}$, $\frac{5}{6}$, $\frac{5}{12}$ **c)** $\frac{4}{5}$, $\frac{1}{2}$, $\frac{3}{10}$

d) $\frac{1}{3}$, $\frac{1}{4}$, $\frac{3}{8}$ **e)** $\frac{2}{3}$, $\frac{5}{9}$, $\frac{11}{18}$ **f)** $\frac{1}{4}$, $\frac{1}{5}$, $\frac{3}{10}$

★ Extension — FOR EARLY FINISHERS

7. Write $\frac{72}{120}$ in its simplest form.

8. Write a fraction that is greater than $\frac{1}{4}$ and less than $\frac{1}{3}$.

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